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Trader Performance vs Bitcoin Market Sentiment

An exploratory analysis of Hyperliquid trading activity
against the Bitcoin Fear & Greed Index

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DATASETS	Bitcoin Fear & Greed Index (2,644 records) Hyperliquid Historical Trades (23,285 records)
PREPARED FOR	PrimeTrade.ai — Data Science Hiring Assignment

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01 Executive Summary

This report investigates how Bitcoin market sentiment — measured via the Fear & Greed Index — relates to trader performance on Hyperliquid, a decentralized derivatives exchange. Trade-level execution records were merged with daily sentiment classifications to uncover patterns in profitability, risk-taking, trading volume, and trader concentration, with the aim of informing sentiment-aware trading strategies.

The analysis shows that sentiment shapes the *shape* of outcomes more than simple averages: extreme gains and losses cluster around Fear and Greed days respectively, overall Buy/Sell activity stays balanced, and a small group of traders captures a disproportionate share of total profit. Volatility has also risen sharply in the most recent period covered by the data, pointing to a shift in market regime.

Total Trades Analyzed	23,285
Sentiment Records	2,644 days
Top Trader Profit	~\$1.6M
Buy / Sell Split	~50.4% / 49.6%

02 Datasets

2.1 Bitcoin Fear & Greed Index

2,644 daily records containing a numeric sentiment score (0–100) and a categorical classification (Extreme Fear, Fear, Neutral, Greed, Extreme Greed), alongside timestamp and date fields.

2.2 Historical Trader Data (Hyperliquid)

23,285 individual trade records including account identifier, traded coin, execution price, trade size (tokens and USD), buy/sell side, position direction, closed PnL, fees, and timestamps.

Column	Description
Account	Trader's wallet / account identifier
Coin	Traded asset (e.g., BTC, ETH, HYPE)
Size USD / Size Tokens	Trade size in USD / tokens
Side / Direction	Buy or Sell; Open/Close Long/Short
Closed PnL	Realized profit or loss on the trade
Fee	Trading fee paid
Timestamp IST	Trade execution time (IST)

03 Methodology

1 Import Libraries

pandas, numpy, matplotlib, and seaborn loaded with clean display settings.

2 Load Datasets

Both CSVs loaded and validated via `.info()` / `.head()` to confirm schema and null counts.

3 Data Cleaning

Parsed Timestamp IST into datetime, derived a common date key, handled minor nulls in Order ID / Fee / Trade ID, removed duplicates, and verified date-range overlap.

4 Merge Datasets

Merged on the common date field so every trade carries its day's sentiment classification and Fear/Greed score.

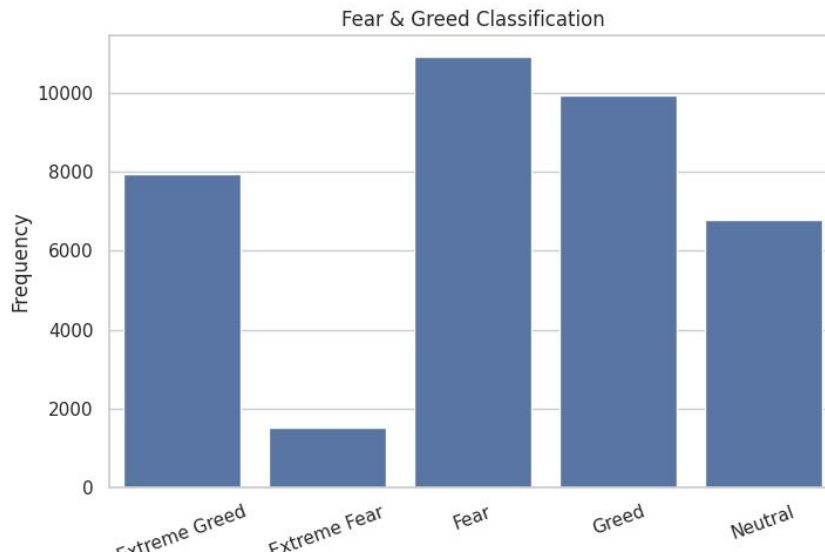
5 Exploratory Analysis

Computed PnL by sentiment, win rate by sentiment, trade size by sentiment, buy/sell mix, most-traded coins, account-level profitability, daily aggregated PnL trend, and cross-feature correlations.

04 Exploratory Data Analysis

The following visualizations summarize trading behavior and profitability patterns across sentiment categories, coins, and time.

4.1 Fear & Greed Classification (Trade-Day Frequency)



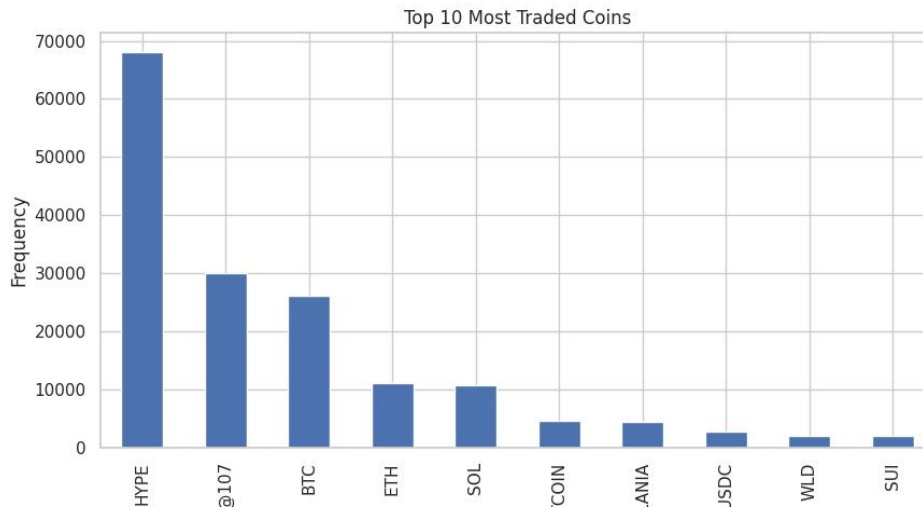
Fear (~10.9K) and Greed (~9.9K) days dominate the dataset's timeline, while Extreme Fear is comparatively rare (~1.5K) — the market spent relatively little time at sentiment extremes over this period.

4.2 Buy vs Sell Trades



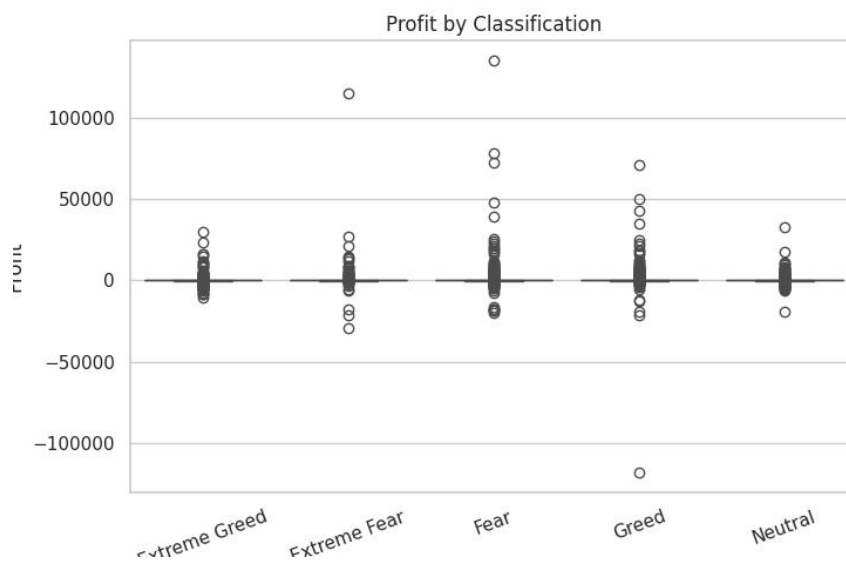
Trade counts are nearly balanced (~18.7K Buy vs ~18.4K Sell), indicating no strong systemic directional bias in this trader cohort.

4.3 Top 10 Most Traded Coins



BTC and ETH lead by trade frequency, with HYPE, SOL, and several mid-cap tokens forming a long tail of activity.

4.4 Profit by Classification



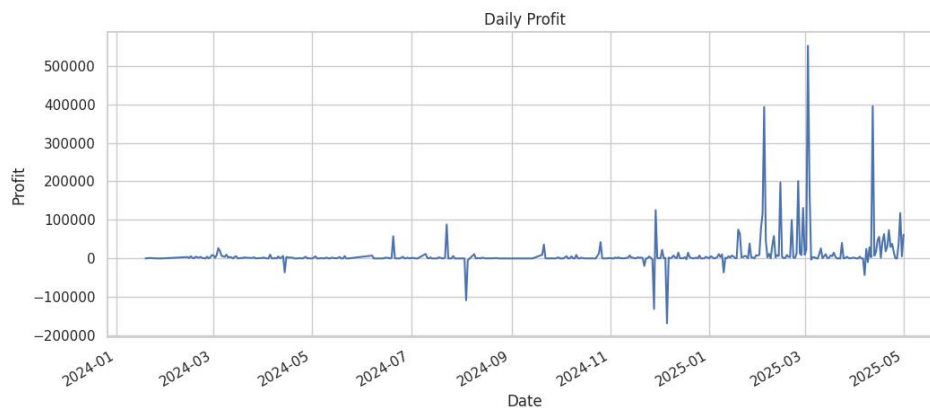
PnL outliers are most extreme during Fear (largest single gain, ~+135K) and Greed (largest single loss, ~-120K) — sentiment extremes coincide with fatter tails in both directions, not uniformly 'safer' or 'riskier' outcomes.

4.5 Profit / Loss Distribution



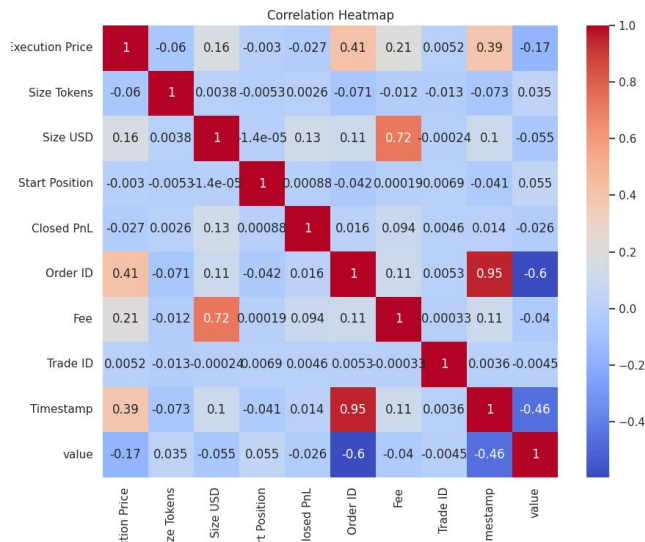
The distribution is heavily concentrated near zero with a long tail on both sides — most trades have small PnL impact, while a few produce very large gains or losses (a classic fat-tailed distribution).

4.6 Daily Aggregate Profit Over Time



Daily PnL volatility increases sharply from around February 2025 onward, with spikes exceeding +500K and drawdowns near -170K, versus a comparatively calm 2024 period — suggesting a shift in market/trading regime.

4.7 Correlation Heatmap



Closed PnL correlates weakly with all other numeric features (max ~ 0.13 with Size USD), suggesting outcomes aren't simply explained by trade size or fees. Size USD and Fee are strongly correlated (0.72), as expected given fee structures. Sentiment value correlates negatively with Order ID / Timestamp (-0.6 / -0.46), reflecting how sentiment trended over the dataset's time span.

4.8 Top 10 Profitable Traders



Profitability is extremely concentrated: the top trader alone earned $\sim 1.6\text{M}$, more than the next two traders combined, and over 20x the 10th-ranked trader ($\sim 60\text{K}$) — a clear power-law / Pareto pattern.

05 Key Insights

- **Sentiment concentration:** Fear and Greed days dominate market activity — the sentiment index rarely sits at Extreme Fear, meaning most trading happens under moderate sentiment conditions rather than panic or euphoria.
- **Asset mix:** BTC and ETH remain the core traded assets, though newer tokens (HYPE, SOL) show meaningful volume, indicating diversification beyond majors.
- **Balanced direction:** Trading volume is close to balanced between Buy and Sell, suggesting no systemic directional bias in this trader cohort.
- **Fat-tailed, sentiment-linked PnL:** The largest gains cluster in Fear periods and the largest single loss occurred during Greed, hinting that extreme moves in either direction coincide with elevated sentiment volatility rather than one sentiment being uniformly 'safer'.
- **Profit concentration:** A small number of accounts, led by one dominant trader (~1.6M), account for a disproportionate share of total profit — a Pareto-style distribution common in trading populations.
- **Rising volatility:** Risk and volatility intensified significantly in 2025, with daily PnL swings growing much larger from February 2025 onward compared to the relatively calm 2024 period.
- **Weak feature correlation:** Closed PnL is not strongly explained by any single numeric feature (size, fee, position) — correlations are all weak, implying trader skill and timing, not just position size, drive outcomes.
- **Clean merge:** No major data quality issues remained after cleaning; the dataset merged cleanly on date with minimal information loss.

06 Business & Trading Implications

- **Risk management:** Since PnL tails are fattest during Fear and Greed (not Neutral), risk limits could be tightened specifically around sentiment extremes rather than applying a flat policy.
- **Strategy design:** The heavy concentration of profit in a handful of accounts suggests studying top-trader behavior (timing, coin selection) could be more valuable than broad-based strategy tweaks.
- **Volatility regime awareness:** The sharp rise in daily PnL swings from 2025 onward signals a regime shift — strategies tuned only on 2024 data may underestimate current risk.
- **Diversification signal:** Emerging tokens (HYPE, SOL) capturing meaningful trade share alongside BTC/ETH suggests sentiment-driven strategies shouldn't be BTC/ETH-only.

07 Conclusion

Combining Hyperliquid trade-level data with the Bitcoin Fear & Greed Index reveals that market sentiment shapes the *shape* of trading outcomes more than it shapes simple averages: extreme profits and losses cluster around Fear and Greed days respectively, overall trading activity is fairly balanced day-to-day, and a small set of traders captures most of the profit pool. Risk and volatility have also grown substantially in the most recent period covered by the data. These patterns suggest that sentiment-aware position sizing and risk controls — tighter limits during high-volatility sentiment regimes, and closer study of the top-performing trader cohort — could meaningfully improve strategy design and risk management.

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